Homework #5

T1 T2 T7

5-1 考虑如图 5-P-1 的微波网络，它包含一个 50 Ω 的源、一个 50 Ω 的 3dB 匹配衰减器和一个 50 Ω 的负载。(a)计算可用功率增益、工作功率增益和传输功率增益；(b)若负载为 25 Ω，增益如何改变？(c)源阻抗为 25 Ω，增益如何改变？

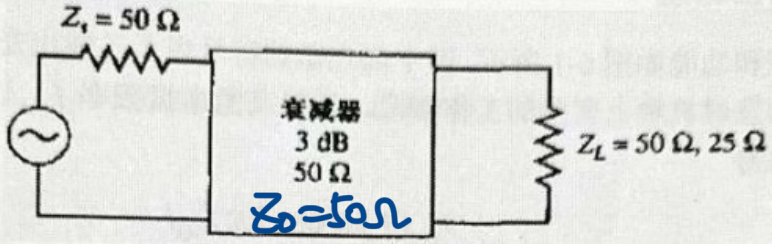


图 5-P-4

解：(a) 3dB衰减器：$S_{11}=S_{22}=0$，$|S_{21}|^2=|S_{12}|^2=10^{-\frac{3}{10}}=0.5$

$Z_S=Z_0 \Rightarrow \Gamma_S=0$，$Z_L=Z_0 \Rightarrow \Gamma_L=0$

$\Gamma_S=\Gamma_L=S_{11}=S_{12}$，系统完全匹配，$P_L=P_{out}$，$P_{avs}=P_{in}$

可用功率增益 $G_A=\frac{P_{avn}}{P_{avs}}=|S_{21}|^2=0.5$

工作功率增益 $G_P=\frac{P_L}{P_{in}}=|S_{21}|^2=0.5$

传输功率增益 $G_T=\frac{P_L}{P_{avs}}=|S_{21}|^2=0.5$

(b) $Z_L=25\Omega$，$Z_S=50\Omega$ 时，$\Gamma_S=0$，$\Gamma_L=\frac{Z_L-Z_0}{Z_L+Z_0}=-\frac{1}{3}$

$\Gamma_{in}=S_{11}+\frac{S_{12}S_{21}\Gamma_L}{1-S_{22}\Gamma_L}=-\frac{1}{6}$　$\Gamma_{out}=S_{22}+\frac{S_{12}S_{21}\Gamma_S}{1-S_{11}\Gamma_S}=0$

$G_A=\frac{1-|\Gamma_S|^2}{|1-S_{11}\Gamma_S|^2}|S_{21}|^2\frac{1}{1-|\Gamma_{out}|^2}=0.5$　$G_P=\frac{1}{1-|\Gamma_{in}|^2}|S_{21}|^2\frac{1-|\Gamma_L|^2}{|1-S_{22}\Gamma_L|^2}=\frac{16}{35}$

$G_T=\frac{1-|\Gamma_S|^2}{|1-\Gamma_{in}\Gamma_S|^2}|S_{21}|^2\frac{1-|\Gamma_L|^2}{|1-\Gamma_{out}\Gamma_L|^2}=\frac{4}{9}$

(c) $Z_L=50\Omega$，$Z_S=25\Omega$ 时，$\Gamma_S=\frac{Z_S-Z_0}{Z_S+Z_0}=-\frac{1}{3}$　$\Gamma_L=0$

$\Gamma_{in}=S_{11}+\frac{S_{12}S_{21}\Gamma_L}{1-S_{22}\Gamma_L}=0$，$\Gamma_{out}=S_{22}+\frac{S_{12}S_{21}\Gamma_S}{1-S_{11}\Gamma_S}=-\frac{1}{6}$

$G_A=\frac{1-|\Gamma_S|^2}{|1-S_{11}\Gamma_S|^2}|S_{21}|^2\frac{1}{1-|\Gamma_{out}|^2}=\frac{16}{35}$　$G_P=\frac{1}{1-|\Gamma_{in}|^2}|S_{21}|^2\frac{1-|\Gamma_L|^2}{|1-S_{22}\Gamma_L|^2}=0.5$

$G_T=\frac{1-|\Gamma_S|^2}{|1-\Gamma_{in}\Gamma_S|^2}|S_{21}|^2\frac{1-|\Gamma_L|^2}{|1-\Gamma_{out}\Gamma_L|^2}=\frac{4}{9}$

5-5 使用 GaAsFET 设计一个在 5GHz 具有最大增益的放大器。系统特征阻抗 50Ω，晶体管 S 参数为：$S_{11}=0.65\angle -140°$，$S_{12}=0.04\angle 60°$，$S_{21}=2.4\angle 50°$，$S_{22}=0.7\angle -65°$。设计出使用开路并联短截线的匹配电路结构。

解，$\Delta = S_{11}S_{22} - S_{12}S_{21} = -0.379 + j0.102 \Rightarrow |\Delta| = 0.392 < 1$

Rollett因子 $K = \frac{1-|S_{11}|^2-|S_{22}|^2+|\Delta|^2}{2|S_{12}S_{21}|} \approx 1.258 > 1$ 故晶体管无条件稳定

$T_S = \left(\frac{B_1 - \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}\right)^* = 0.325\angle -147.1°$

$T_L = \left(\frac{B_2 - \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}\right)^* = 0.843\angle -70.5°$

$y_{in} = \frac{1-\Gamma_{in}}{1+\Gamma_{in}} = \frac{1-\Gamma_S^*}{1+\Gamma_S^*} = 1.08 - j3.04$

$y_{out} = \frac{1-\Gamma_{out}}{1+\Gamma_{out}} = \frac{1-\Gamma_L^*}{1+\Gamma_L} = 0.17 - j0.699$

$\lambda = \frac{c}{f} = 0.06m$

① 对输入匹配

求得 $\beta d = 0.737° \Rightarrow d = 0.002\lambda = 0.12mm$

$\beta l = 71.68° \Rightarrow l_{in} = 0.199\lambda = 11.94mm$

② 对输出匹配

求得 $\beta d = 109.0° \Rightarrow d = 0.303\lambda = 18.18mm$

$\beta l = 107.7° \Rightarrow l_{out} = 0.299\lambda = 17.94mm$

5-7 一个 GaAsFET 在 8GHz 具有如下参数：$S_{11}=0.7\angle-110°$，$S_{12}=0.02\angle60°$，$S_{21}=3.5\angle60°$，$S_{22}=0.8\angle-70°$，$F_{min}=2.5dB$，$\Gamma_{opt}=0.7\angle120°$，$r_n=15\Omega$。使用开路并联短截线的匹配电路结构，设计具有最小噪声系数和最大可能增益的放大器。

解：$\lambda=\frac{c}{f}=0.0375m$

① 最小噪声系数

$\Gamma_S=\Gamma_{opt}=0.7\angle120°$

$y_S=\frac{1-\Gamma_S}{1+\Gamma_S}=0.647-j1.535$

求得 $\beta d=172.82°$

$\Rightarrow d_{in}=0.430\lambda=$ 18mm

$\beta l_{in}=62.96°$

$\Rightarrow l_{in}=0.175\lambda=$ 6.562mm

② 最大可能增益

$\Gamma_L=\Gamma_{out}=S_{22}+\frac{S_{12}S_{21}\Gamma_S}{1-S_{11}\Gamma_S}=0.873\angle-74.0°$

$y_L=\frac{1-\Gamma_L}{1+\Gamma_L}=0.106-j0.748$

求得 $\beta d=141.62°$

$\Rightarrow d_{out}=0.393\lambda=$ 14.737mm

$\beta l_{out}=74.38°$

$l_{out}=0.207\lambda=$ 7.762mm